

**LAB 5: Create a php based Application and Launch it  
using Beanstalk**

**Lab Report Submitted to the**

**DR.Hariharasitaraman , VIT BHOPAL UNIVERSITY**

**BACHELOR OF TECHNOLOGY  
COMPUTER SCIENCE ENGINEERING**

**Name - Arpita Rajput  
Reg. No. - 22MIP10001**



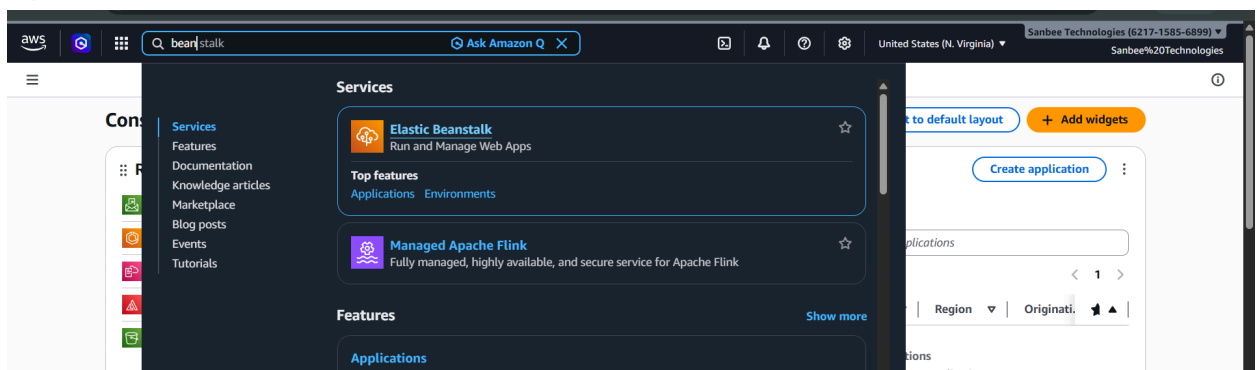
**School of Computer Science Engineering  
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**Feb, 2026**

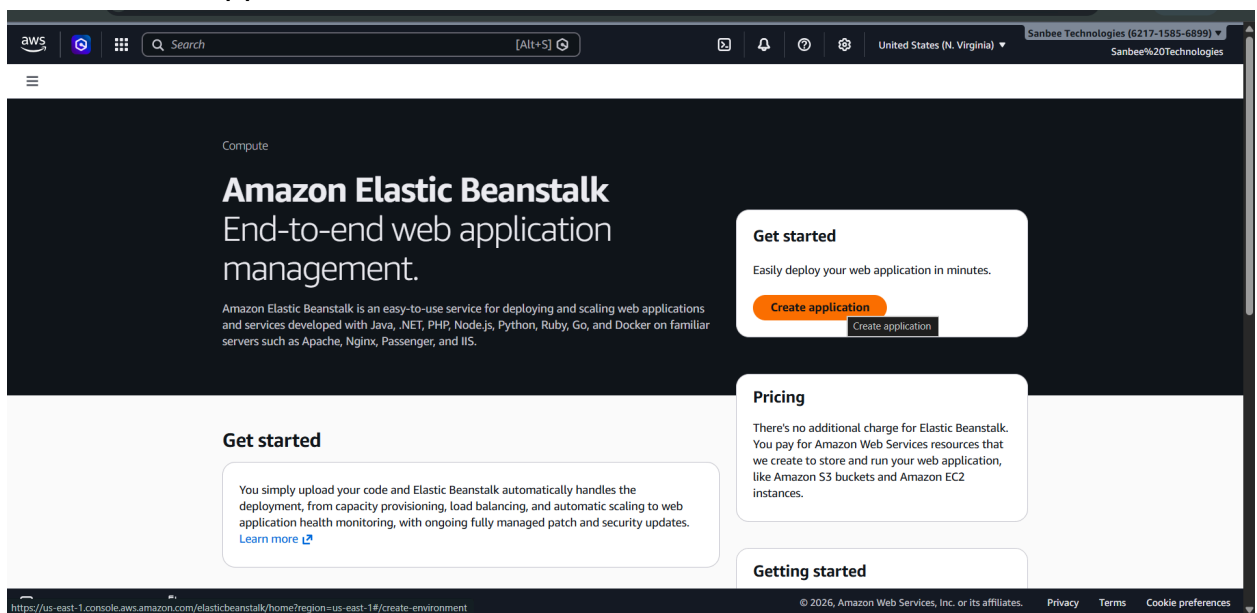
# AIM: Create a PHP-based Application and Launch it using AWS Elastic Beanstalk — Steps Only

## Step 1: Create Application

1. Open AWS Elastic Beanstalk console.



2. Click Create Application.



### 3. Enter application name.

The screenshot shows the 'Configure environment' step in the AWS Elastic Beanstalk console. On the left, a vertical list of steps is shown: Step 1 (Configure environment), Step 2 (Configure service access), Step 3 (optional: Set up networking, database, and tags), Step 4 (optional: Configure instance traffic and scaling), Step 5 (optional: Configure updates, monitoring, and logging), and Step 6 (Review). The main area is titled 'Configure environment' and contains two sections: 'Environment tier' and 'Application information'. In the 'Environment tier' section, 'Web server environment' is selected. In the 'Application information' section, the 'Application name' field contains '22MIP10001\_BS'. A 'Switch to new version' button is visible in the top right corner of the console area.

**Configure environment**

**Environment tier**

Amazon Elastic Beanstalk has two types of environment tiers to support different types of web applications.

- ☒ **Web server environment**  
Run a website, web application, or web API that serves HTTP requests. [Learn more](#)
- ☐ **Worker environment**  
Run a worker application that processes long-running workloads on demand or performs tasks on a schedule. [Learn more](#)

**Application information**

**Application name**

22MIP10001\_BS

Maximum length of 100 characters.

**Application tags (optional)**

### 4. Select PHP platform.

The screenshot shows the 'Platform' selection step in the AWS Elastic Beanstalk console. The 'Platform' dropdown is set to 'PHP'. The 'Platform branch' dropdown is set to 'PHP 8.5 running on 64bit Amazon Linux 2023'. The 'Platform version' dropdown is set to '4.9.3 (Recommended)'. The 'Application code' section shows 'Sample application' selected. The 'Presets' section shows 'Single instance (free tier eligible)' selected. The 'Environment description' section is collapsed.

**Platform**

Platform: PHP

Platform branch: PHP 8.5 running on 64bit Amazon Linux 2023

Platform version: 4.9.3 (Recommended)

**Application code**

- ☒ **Sample application**  
Existing version  
Application versions that you have uploaded.
- ☐ **Upload your code**  
Upload a source bundle from your computer or copy one from Amazon S3.

**Presets**

Start from a preset that matches your use case or choose custom configuration to unset recommended values and use the service's default values.

**Configuration presets**

- ☒ **Single instance (free tier eligible)**

### 5. Create environment and launch application.

## Configure environment [Info](#)

### Environment tier [Info](#)

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Elastic Beanstalk

> Create environment

1

You're using a legacy workflow

AWS Elastic Beanstalk recently released a new console workflow that makes it easier to use. You can continue to use the legacy workflow until it is deprecated.

Switch to new version

Step 1

Configure environment

Step 2

**Configure service access**

Step 3 - optional

Set up networking, database, and tags

Step 4 - optional

Configure instance traffic and scaling

Step 5 - optional

Configure updates, monitoring, and logging

Step 6

Review

**Configure service access**

[Info](#)

**Service access**

IAM roles, assumed by Elastic Beanstalk as a service role, and EC2 instance profiles allow Elastic Beanstalk to create and manage your environment. Both the IAM role and instance profile must be attached to IAM managed policies that contain the required permissions. [Learn more](#)

**Service role**

Choose an IAM role for Elastic Beanstalk to assume as a service role. The IAM role must have the required IAM managed policies.

aws-elasticbeanstalk-service-role

Create role

**EC2 instance profile**

Choose an IAM instance profile with managed policies that allow your EC2 instances to perform required operations.

aws-elasticbeanstalk-ec2-role

Create role

**EC2 key pair - optional**

Select an EC2 key pair to securely log in to your EC2 instances. [Learn more](#)

Choose a key pair

Cancel

Skip to Review

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Elastic Beanstalk

> Create environment

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Configure service access

Step 3 - optional

Set up networking, database, and tags

Step 4 - optional

Configure instance traffic and scaling

Step 5 - optional

Configure updates, monitoring, and logging

Step 6

**Review**

**Review**

[Info](#)

**Step 1: Configure environment**

[Edit](#)

**Environment information**

**Environment tier**

Web server environment

**Environment name**

22MIP10001BS-env

**Platform**

arn:aws:elasticbeanstalk:us-east-1::platform/PHP 8.5 running on 64bit Amazon Linux 2023/4.9.3

**Application name**

22MIP10001\_BS

**Application code**

Sample application

**Step 2: Configure service access**

[Edit](#)

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Elastic Beanstalk > Create environment

Step 2: Configure service access

Edit

Service access

Info

Configure the service role and EC2 instance profile that Elastic Beanstalk uses to manage your environment. Choose an EC2 key pair to securely log in to your EC2 instances.

Service role

arn:aws:iam::621715856899:role/aws-elasticbeanstalk-service-role

EC2 instance profile

aws-elasticbeanstalk-ec2-role

Step 3: Set up networking, database, and tags

Edit

Networking, database, and tags

info

Configure VPC settings, and subnets for your environment's EC2 instances and load balancer. Set up an Amazon RDS database that's integrated with your environment.

No options configured

Tags

| Key                       | Value |
|---------------------------|-------|
| No tags                   |       |
| There are no tags defined |       |

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Elastic Beanstalk > Create environment

Step 4: Configure instance traffic and scaling

Edit

Instance traffic and scaling

Info

Customize the capacity and scaling for your environment's instances. Select security groups to control instance traffic. Configure the software that runs on your environment's instances by setting platform-specific options.

Instances

IMDSv1

Disabled

Capacity

Environment type

Single instance

On-demand above base

70

Processor type

x86\_64

Fleet composition

On-Demand instance

Capacity rebalancing

Disabled

Instance types

t3.micro,t3.small

On-demand base

0

Scaling cooldown

360

AMI ID

ami-00039f9fc3a4ad4fc

Step 5: Configure updates, monitoring, and logging

Edit

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Elastic Beanstalk > Create environment

Step 5: Configure updates, monitoring, and logging

Edit

Updates, monitoring, and logging

Define when and how Elastic Beanstalk deploys changes to your environment. Manage your application's monitoring and logging settings, instances, and other environment resources.

Monitoring

|                           |                                      |                                         |
|---------------------------|--------------------------------------|-----------------------------------------|
| System enhanced           | Cloudwatch custom metrics - instance | Cloudwatch custom metrics - environment |
| Log streaming Disabled    | Retention 7                          | Lifecycle false                         |
| Updates                   |                                      |                                         |
| Managed updates Enabled   | Deployment batch size 100            | Deployment batch size type Percentage   |
| Command timeout 600       | Deployment policy AllAtOnce          | Health threshold Ok                     |
| Ignore health check false | Instance replacement false           |                                         |
| Platform software         |                                      |                                         |
| Lifecycle                 | Log streaming                        | Allow URL fopen                         |

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us-east-1.console.aws.amazon.com/elasticbeanstalk/home?region=us-east-1#/create-environment

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Elastic Beanstalk > Create environment

|                        |                             |                       |
|------------------------|-----------------------------|-----------------------|
| Lifecycle false        | Log streaming Disabled      | Allow URL fopen On    |
| Display errors Off     | Document root -             | Max execution time 60 |
| Memory limit 256M      | Zlib output compression Off | Proxy server nginx    |
| Logs retention 7       | Rotate logs Disabled        | Update level minor    |
| X-Ray enabled Disabled |                             |                       |

Environment properties

Source

Key

Value

No environment properties  
There are no environment properties defined

Cancel Previous Create

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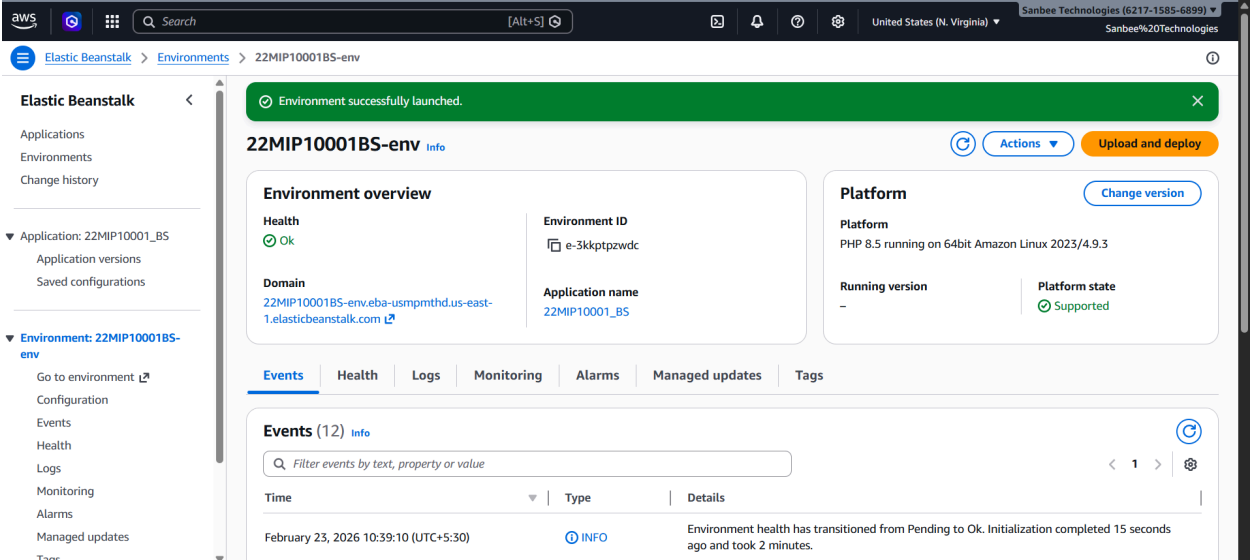
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## Step 2: Deploy Application

1. Elastic Beanstalk automatically deploys the application.
2. Wait 5 minutes for resources to be created.

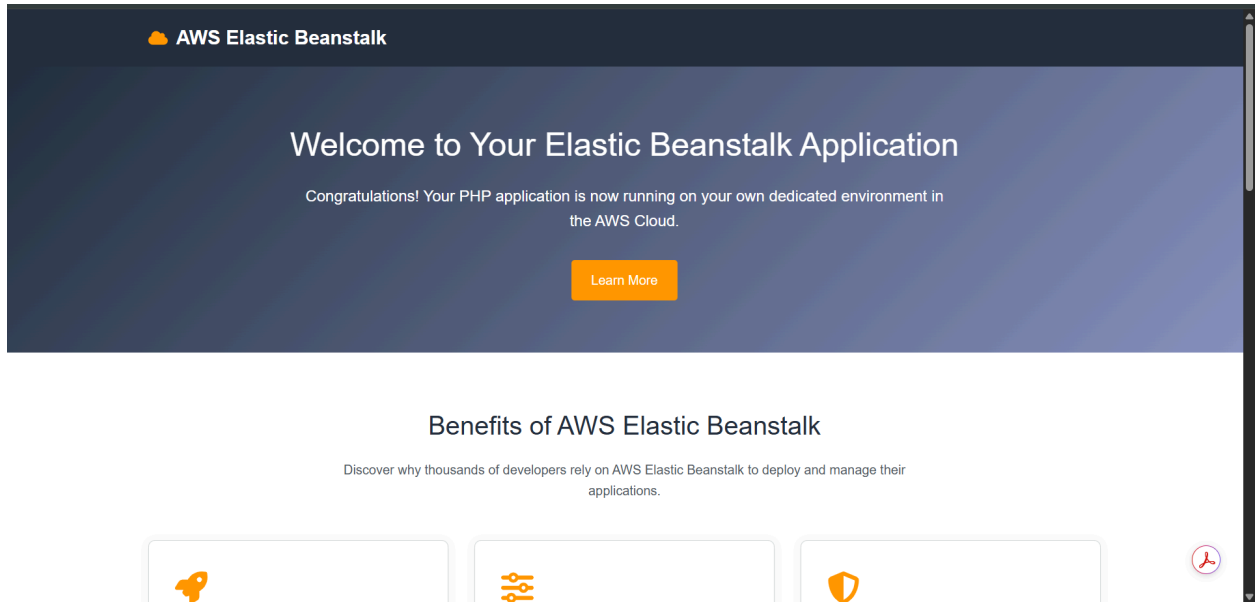
3.



The screenshot shows the AWS Elastic Beanstalk console. At the top, a green banner states "Environment successfully launched." Below this, the environment name "22MIP10001BS-env" is displayed. The "Environment overview" section shows the health status as "Ok" (green checkmark). The "Platform" section indicates "PHP 8.5 running on 64bit Amazon Linux 2023/4.9.3". The "Events" section shows a list of events, with the most recent event stating "Environment health has transitioned from Pending to Ok. Initialization completed 15 seconds ago and took 2 minutes." The left sidebar shows the navigation menu with "Environments" selected. The top navigation bar shows the AWS logo, search bar, and account information.

## Step 3: Explore Environment

1. Open Elastic Beanstalk console → Select Region.
2. Go to Environments → Select environment name.
3. Click Go to Environment or use domain URL to view app.



## Step 4: Troubleshoot Using Logs

1. Open environment → Logs tab.
2. Click Request Logs → Select Last 100 lines.

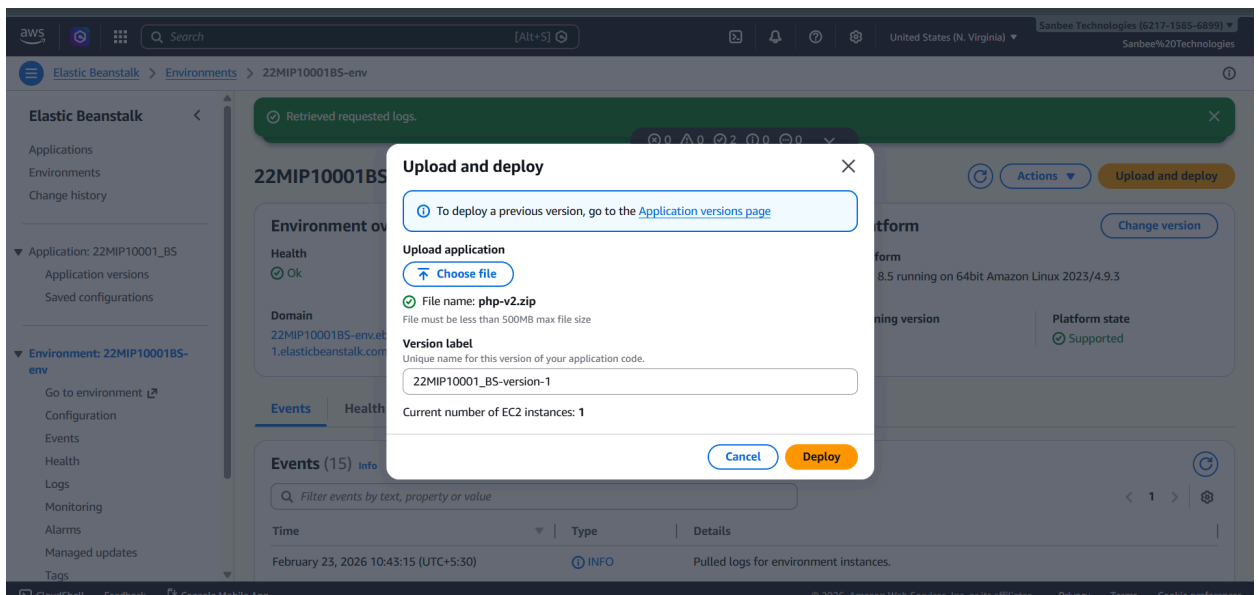


### 3. Download logs and check nginx access log.

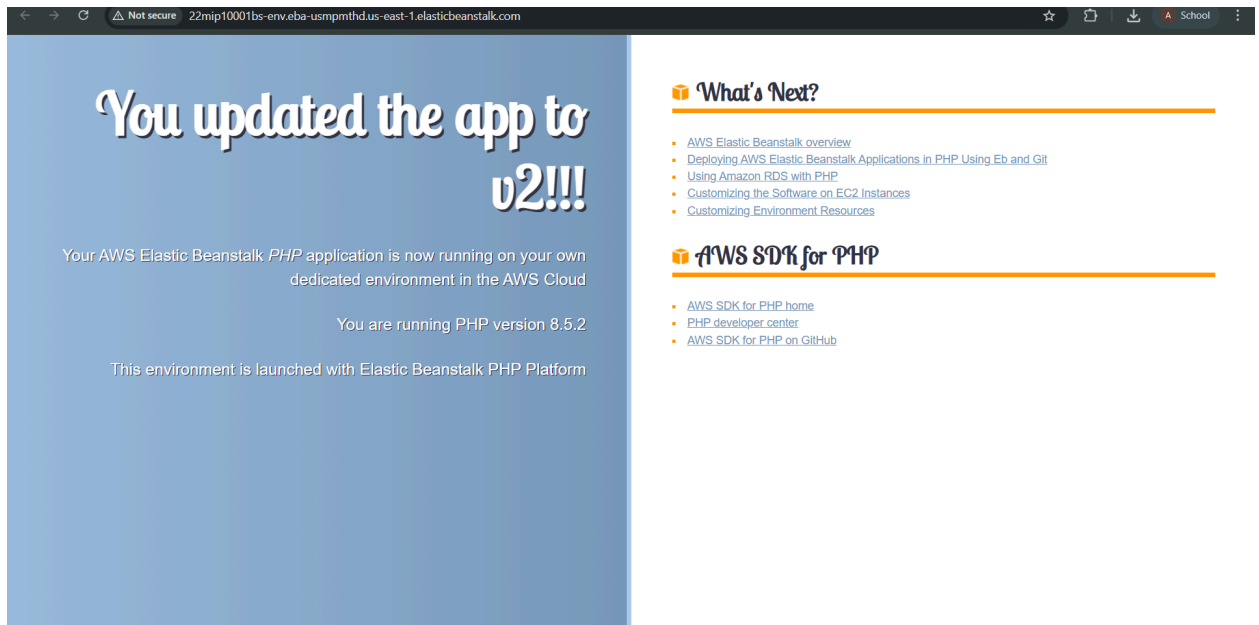
```
-----  
/var/log/eb-hooks.log  
-----  
  
/var/log/nginx/access.log  
139.167.143.182 - - [23/Feb/2026:05:10:59 +0000] "GET / HTTP/1.1" 200 20329 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/145.0.0.0 Safari/537.36" "-"  
34.22.177.31 - - [23/Feb/2026:05:11:04 +0000] "GET / HTTP/1.1" 200 20329 "-" "python-requests/2.32.5" "-"  
  
/var/log/nginx/error.log  
2026/02/23 05:08:36 [warn] 2044#2044: could not build optimal types_hash, you should increase either types_hash_max_size: 1024 or types_hash_bucket_size: 64; ignoring types_hash_bucket_size  
2026/02/23 05:08:37 [warn] 2077#2077: could not build optimal types_hash, you should increase either types_hash_max_size: 1024 or types_hash_bucket_size: 64; ignoring types_hash_bucket_size  
2026/02/23 05:08:37 [warn] 2078#2078: could not build optimal types_hash, you should increase either types_hash_max_size: 1024 or types_hash_bucket_size: 64; ignoring types_hash_bucket_size  
  
/var/log/eb-engine.log  
2026/02/23 05:08:33.570403 [INFO] disable log stream  
2026/02/23 05:08:33.570415 [INFO] Running command: systemctl show -p PartOf amazon-cloudwatch-agent.service  
2026/02/23 05:08:33.587051 [INFO] Running command: systemctl stop amazon-cloudwatch-agent.service  
2026/02/23 05:08:33.603288 [INFO] Executing instruction: GetToggleForceRotate  
2026/02/23 05:08:33.603313 [INFO] Checking if logs need forced rotation  
2026/02/23 05:08:33.603326 [INFO] Running command: /opt/aws/bin/cfn-get-metadata -s arn:aws:cloudformation:us-east-1:621715856899:stack/awseb-e-3kktptzdc-stack/69929c30-1075-11f1-bf7f-0e811aa287f7 -r  
AWSBatchScalingGroup --region us-east-1  
2026/02/23 05:08:33.977580 [INFO] Running command: /opt/aws/bin/cfn-get-metadata -s arn:aws:cloudformation:us-east-1:621715856899:stack/awseb-e-3kktptzdc-stack/69929c30-1075-11f1-bf7f-0e811aa287f7 -r  
AWSBatchScalingGroup --region us-east-1  
2026/02/23 05:08:34.377105 [INFO] Profile not found. Generating default rsyslog config  
2026/02/23 05:08:34.379518 [INFO] Running command: systemctl restart rsyslog.service  
2026/02/23 05:08:34.947432 [INFO] Executing instruction: PostBuildEbExtension  
2026/02/23 05:08:34.947520 [INFO] Starting executing the config set Infra-EmbeddedPostBuild.  
2026/02/23 05:08:34.947534 [INFO] Running command: /opt/aws/bin/cfn-init -s arn:aws:cloudformation:us-east-1:621715856899:stack/awseb-e-3kktptzdc-stack/69929c30-1075-11f1-bf7f-0e811aa287f7 -r  
AWSBatchScalingGroup --region us-east-1 --configsets Infra-EmbeddedPostBuild  
2026/02/23 05:08:35.274177 [INFO] Finished executing the config set Infra-EmbeddedPostBuild.  
  
2026/02/23 05:08:35.274200 [INFO] Executing instruction: CleanEbExtensions  
2026/02/23 05:08:35.274323 [INFO] Cleaned extensions subdirectories from app staging directory.  
2026/02/23 05:08:35.274327 [INFO] Executing instruction: RunAppDeployPreDeployHooks  
2026/02/23 05:08:35.274341 [INFO] Executing platform hooks in .platform/hooks/predeploy/  
2026/02/23 05:08:35.274352 [INFO] The dir .platform/hooks/predeploy/ does not exist  
2026/02/23 05:08:35.274355 [INFO] Finished running scripts in /var/app/staging/.platform/hooks/predeploy  
2026/02/23 05:08:35.274365 [INFO] Executing instruction: stop X-Ray
```

## Step 5: Update Application

1. Download PHP sample app (php-v2.zip).
2. Open environment → Click Upload and Deploy.

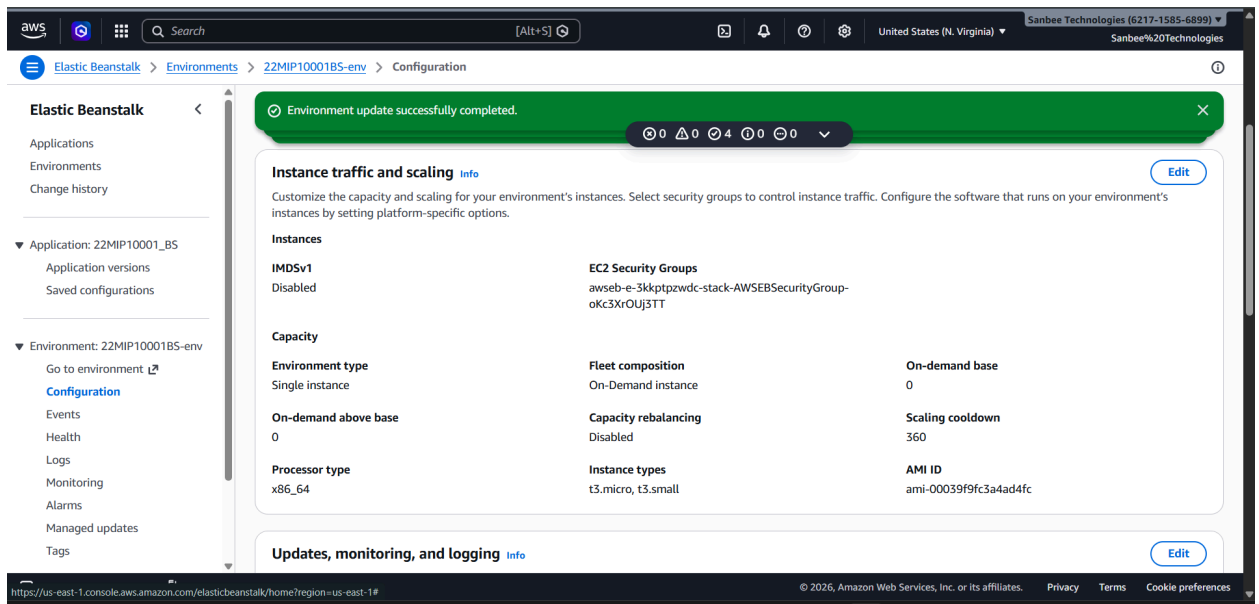


### 3. Upload file → Choose Deploy.



## Step 6: Scale Application

1. Open environment → Configuration.
2. Edit Instance Traffic and Scaling.



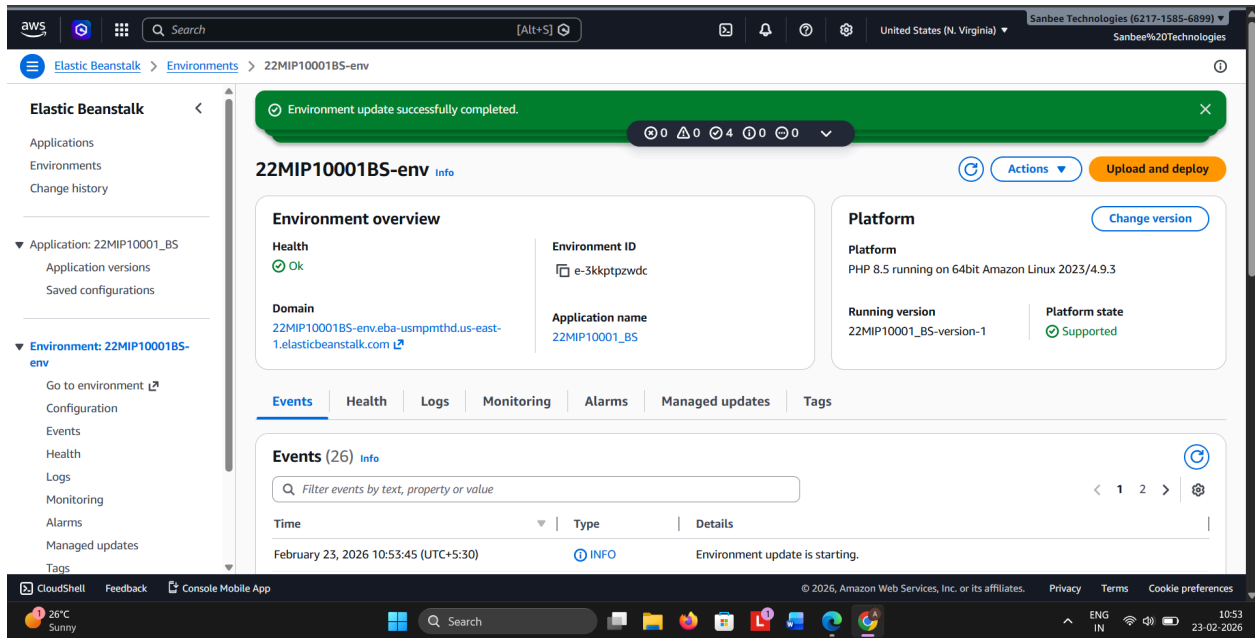
3. Change environment type to Load Balanced.

4. Set Min instances = 2, Max = 4.

5. Click Apply.

The screenshot shows the AWS Elastic Beanstalk Configuration page for environment `22MIP10001B5-env`. A green banner at the top indicates "Environment update successfully completed." Below this, the "Capacity" section is expanded, showing the "Auto scaling group" configuration. The "Environment type" is set to "Load balanced". The "Min instances" is set to 2 and the "Max instances" is set to 4. Under "Fleet composition", "On-Demand instances" is selected. The "Instance types" list shows "t3.micro" as the first instance type. The bottom of the page shows the "CloudShell", "Feedback", and "Console Mobile App" links, along with the copyright notice "© 2026, Amazon Web Services, Inc. or its affiliates."

The screenshot shows the "Listener rules" section of the AWS Elastic Beanstalk Configuration page. A green banner at the top indicates "Environment update successfully completed." Below this, a table lists listener rules, but it is empty with the message "No additional listener rules are currently configured. Click Add rule to add a listener rule." Below the table, the "Log files access" section is expanded, showing the "Store logs" option. The "Store logs" checkbox is currently unchecked. The bottom of the page shows the "Cancel", "Continue", and "Apply" buttons, along with the copyright notice "© 2026, Amazon Web Services, Inc. or its affiliates."



## Step 7: Clean Up Resources

1. Delete application versions from console and Amazon S3.
2. Terminate environment.
3. Delete application.